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Review

Untying the knot of behavioural rebound, negative spillover, and crowding-out mechanisms in environmental sustainability research: a systematic literature review

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HIGHLIGHTS

- Behavioural rebound, negative spillover, and crowding-out are used inconsistently.
- The concepts describe behavioural mechanisms that offset environmental gains.
- The mechanisms differ on a structural and contextual level.
- The mechanisms have different input, output, and overall mechanism characteristics.
- Recommendations for future studies facilitate interpretation and integration.

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ABSTRACT

A comprehensive understanding of the mechanisms underlying rebound effects is crucial for explaining the observed offset in environmental gains. Nevertheless, research on behavioural rebound mechanisms remains scarce and is characterised by inconsistent terminology. The terms "behavioural rebound", "negative spillover", and "crowding-out" are used interchangeably, yet understanding of their differences and similarities remains lacking. To address this gap, a systematic literature review has been conducted, identifying 29 behavioural rebound, 30 negative spillover, and 20 crowding-out mechanisms, derived from 52 articles. The results highlight that the mechanisms differ on a structural (e.g., causal elements) and contextual level (e.g., analysed applications) in terms of input (e.g., presence of external intervention and intervention characteristics), output (e.g., identified endpoint and domain) and overall mechanism characteristics (e.g., number of behavioural changes and described effect type). This indicates that although behavioural rebound, negative spillover, and crowding-out mechanisms describe overlapping phenomena, the concepts should not be used as synonyms. Alternatively, negative spillover mechanisms can be considered as a type of behavioural rebound mechanisms. Furthermore, the results imply that crowding-out mechanisms could lead to negative spillover mechanisms, which in turn could lead to behavioural rebound mechanisms. This study concludes with recommendations for future research to improve interpretation as well as integration of the different concepts, supporting research on the often neglected but highly relevant behavioural mechanisms underlying rebound effects.

1. Introduction

To be able to successfully tackle the urgent environmental sustainability challenges, it is crucial to understand and prevent rebound effects. Rebound effects occur when the potential environmental benefits of sustainability-oriented interventions are offset due to systemic or behavioural responses (Hertwich, 2005; Lange et al., 2021). Previous studies indicate that rebound effects offset ca. 40 % of the potential environmental gains and, in some cases, even lead to a negative net impact (i.e., backfire) (Antal & Van Den Bergh, 2013; Chang et al., 2018; Chitnis and Sorrell, 2015; Santarius and Soland, 2018).

The majority of research on rebound effects has examined the underlying mechanisms from a traditional economic perspective, often explaining rebound effects as the result of a change in relative prices or

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available budget while assuming rational decision processes (Chitnis and Sorrell, 2015; Gillingham et al., 2016). Behavioural studies demonstrate that rebound effects are additionally influenced by psychological and social factors such as moral credits, intrinsic motivation, and cognitive biases (Santarius and Soland, 2018; Van der Loo and Pigosso, 2024). However, a comprehensive behavioural understanding of rebound effects remains lacking.

Currently, different concepts are used to explain the behavioural mechanisms underlying rebound effects, including 'behavioural rebound', 'negative spillover', and 'crowding-out'. Nevertheless, it remains unclear whether the different concepts represent the same or distinct phenomena and, if different, on what grounds they can be differentiated (Kumar et al., 2023; Reimers et al., 2021; Truelove et al., 2014). The lack of a clear and integrated framework on behavioural rebound, negative spillover, and crowding-out mechanisms limits the interpretation and integration of findings and further delays the development of a comprehensive behavioural understanding of rebound effects (Bracken and Oughton, 2006; Pischke et al., 2017).

To overcome this barrier, this study employs a systematic literature review to provide a thorough comparison between behavioural rebound, negative spillover, and crowding-out mechanisms triggered by sustainability-oriented interventions, along with recommendations for future research.

1.1. Background literature

1.1.1. Rebound effects

The definition of rebound effects can be traced back to Wiliam Stanley Jevon. Already in 1865, he argued that increased resource efficiency leads to an increase rather than decrease in the overall consumption of resources, also known as Jevon's paradox (Alcott, 2005; Jevons, 2023). Currently, rebound effects are defined in more general terms, for example, as an offset in the potential sustainability benefits of an intervention due to systemic or behavioural responses (Hertwich, 2005; Lange et al., 2021). Rebound effects are regularly classified into direct, indirect, and economy-wide effects (Brockway et al., 2021; Saunders, 2013). Whereas direct rebound effects refer to an increase in resource consumption for the same product or service, indirect rebound effects refer to increased consumption of other products or services. The combination of direct and indirect rebound effects can lead to the economy-wide rebound effect, referring to a rebalance in the economic system (Brockway et al., 2021).

Research on rebound effects has mostly been conducted in the field of environmental sustainability, nevertheless, a few exceptions include the contexts of audit reviewing (Frank and Hoffman, 2015), social judgments (Geeraert and Yzerbyt, 2007), and appraisal suppression (Yap and Tong, 2009). Within the context of environmental sustainability, certain domains have been researched more extensively than others. For example, while the majority of rebound effect studies predominantly focuses on the energy domain, other sustainability dimensions such as water pollution and land use have remained comparatively underexplored (Andrew and Pigosso, 2024). Similarly, interventions targeting efficiency (i.e., reducing the resources needed for the same service or function (van den Bergh, 2011)) have received disproportionate more attention in research than interventions targeting effectiveness (i.e., extending the life cycle of resources (Braungart et al., 2007)) and sufficiency (i.e., mitigating the demand of resource consumption (Bocken and Short, 2016; Andrew and Pigosso, 2024).

Mainstream economists frequently argue that rebound effects occur due to an intervention decreasing the costs of consuming a product or service, which will lead to additional available budget, ultimately resulting in the consumption of more units of the same product or service (i.e., income effect, direct rebound) or other products or services (i.e., substitution effect, indirect rebound) (Guzzo et al., 2024; Metic and Pigosso, 2022). Nevertheless, it is becoming widely accepted that including social and psychological perspectives is of high importance for a comprehensive understanding of rebound effects (Santarius and Soland, 2018; Van der Loo and Pigosso, 2024). From a behavioural perspective, rebound effects, also referred to as behavioural rebound effects (e.g., Dorner, 2019), are often explained as the result of moral licensing, indicating that the performance of an initial moral behaviour increases the moral credits or credentials, which creates the perception of entitlement to perform subsequent morally questionable behaviour or inaction (Dütschke et al., 2018; Nash et al., 2017; Nilsson et al., 2017). Even though less common, a variety of other high potential behavioural mechanisms underlying rebound effects, such as diffusion of responsibility (see section 3.1.), have been addressed and are expected to receive more attention in the future (Santarius and Soland, 2018; Van der Loo and Pigosso, 2024).

1.1.2. Spillover effects

The spillover effect demonstrates how engaging in one behaviour influences the likelihood or extent of engaging in other behaviours (Lanzini and Thøgersen, 2014; Nilsson et al., 2017). The spillover effect can be further categorized in several manners. For example, spillover effects can be behavioural (i.e., different behaviours), temporal (i.e., the same behaviour in different moments in time), or contextual (i.e., the same behaviour in different contexts) (Nilsson et al., 2017). Similarly, spillover effects can be promoting (i.e., a desired behaviour leading to a desired behaviour), permitting (i.e., a desired behaviour leading to an undesired behaviour), purging (i.e., an undesired behaviour leading to a desired behaviour), and precipitating (i.e., an undesired behaviour leading to an undesired behaviour) (Dolan and Galizzi, 2015; Galizzi and Whitmarsh, 2019).

Spillover effects are either positive or negative (Geiger et al., 2021; Kumar et al., 2023; Maki et al., 2019). Applied to the context of environmental sustainability, negative spillovers occur when engagement in one pro-environmental behaviour decreases or prevents engagement in other pro-environmental behaviours (Truelove et al., 2014). In addition to moral licensing (Carrico, 2021; Nilsson et al., 2017), the single action bias (i.e., limited subsequent pro-environmental behaviour due to perceived goal fulfilment) and contribution ethics (i.e., limited subsequent pro-environmental behaviour due to the perception of appropriate contribution) (Nash et al., 2017; Truelove et al., 2014) have been used to explain the occurrence of negative spillover effects.

On the contrary, positive spillover effects occur when engaging in one pro-environmental behaviour promotes engagement in subsequent pro-environmental behaviours (Truelove et al., 2014). This phenomenon is often explained by the self-identity theory, indicating that performing a pro-environmental behaviour may lead people to experience a stronger environmental self-identity and therefore also greater moral obligation to continue acting environmentally friendly (Lacasse, 2016; Bern, 1972; Nilsson et al., 2017). Another commonly used explanation is peoples need for consistency, predicting that positive spillover effects arise because people are motivated to behave consistently across times and contexts (Festinger, 1957; Nilsson et al., 2017).

In addition to pro-environmental behaviours, spillovers effects have been applied to a variety of behaviours in different contexts (Galizzi and Whitmarsh, 2019), such as donating to charity (e.g., Alt and Gallier, 2022), volunteering (e.g., Dolan and Galizzi, 2015), unhealthy diets (e.g., Inzlicht and Kang, 2010), and logo liking (e.g., Bettels and Wiedmann, 2019). Moreover, in addition to behavioural mechanisms, some studies have identified spillovers based on changes in culture (e.g., Hogben et al., 2001), knowledge (e.g., Jerome, 2013), and policy (e.g., Tule et al., 2019).

1.1.3. Crowding effects

A frequently cited example of the crowding-out effect originates from Titmuss (1971), who demonstrated that offering monetary compensation for blood donation could paradoxically reduce, rather than increase, the number of donors (Mellström and Johannesson, 2008).

Since then, the crowding-out effect has been applied to a variety of different contexts, such as nuclear energy facilitation (Frey and Oberholzer-Gee, 1997), daycare arrival times (Gneezy and Rustichini, 2000), biodiversity and ecosystem conservation (Rode et al., 2015), and policy support for climate change (Knook et al., 2022). The crowding-out effect mostly describes how an external intervention (e.g., commands or incentives) can undermine the desired performance (Rode et al., 2015). Nevertheless, exceptions exist including the examination of the effect of household behaviour instead of external interventions (Werfel, 2017). Even though less often examined, crowding-in effects are possible, which occur when an external intervention reinforces rather than undermines the desired performance (Rode et al., 2015).

From a behavioural perspective, crowding-out is often used interchangeably with motivational crowding, which is based on the concept of extrinsic motivations (i.e., relying on external pressure) and intrinsic motivations (i.e., existing within the individual) (Ryan and Deci, 1985). Motivational crowding hypothesizes that crowding-out occurs when external incentives overrule people's intrinsic motivations because of, for instance, control aversion or identity changes (Lee, 2022; Frey, 1992; Rode et al., 2015). From an economic perspective, the motivational crowding-out hypothesis was initially met with scepticism, as standard economic theory would predict that any additional reward for a given activity could only increase an individual's motivation to perform that activity (Lacetera and Macis, 2010; Lee, 2022; Mellström and Johannesson, 2008). Nevertheless, the concept of motivational crowding is becoming increasingly accepted. In addition to motivational crowding, several economic explanations have been identified for crowding-out. One example is the hard to get mechanism, indicating that the potential to earn rewards in return for an activity may create incentives to not perform the targeted activity in an attempt to increase those rewards (Schnedler and Vanberg, 2014).

1.1.4. Cross-concept analysis: rebound, spillover and crowding effects

Despite the relatively clear distinct definitions of rebound, spillover, and crowding effects, confusion on the similarities and differences arises when comparing their behavioural explanations for offsetting effects (i.e., behavioural rebound, negative spillover, and crowding-out mechanisms), in the context of environmental sustainability (Fig. 1.).

This confusion can be observed when comparing the concept of behavioural rebound and negative spillover. Whereas several studies argue that the two concepts have the same meaning (e.g., Isbanner et al., 2021; Knook et al., 2022), other studies argue that the two concepts are different but disagree on what grounds. For instance, Reimers et al. (2021, 2022) argue that spillover effects lead to rebound effects, while Dreijerink et al. (2021) argues the other way around stating that rebound effects lead to spillover effects. Similarly, Nash et al. (2017) argues that rebound and spillover effects occur in parallel, whereas Dütschke et al. (2018) and Nilsson et al. (2017) argue that rebound effects are a type of spillover effects.

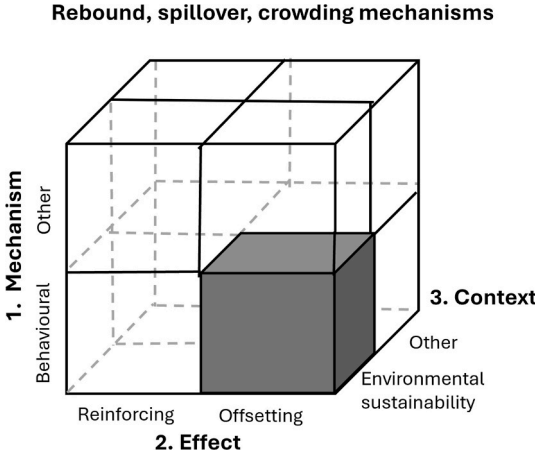


Fig. 1. Visual representation of the scope of this study.

Even though the comparison between the concept of negative spillover and crowding-out occurs less frequently, similar confusion exists. Several studies mention that the two concepts are similar (e.g., Ek, 2018; Knook et al., 2022; Lacroix et al., 2022; Werfel, 2017). Nevertheless, Truelove et al. (2014) and Clot et al. (2022) argue differently stating that crowding-out effects can lead to negative spillover effects. Lastly, apart from the study of Knook et al. (2022), which briefly mentions that all three concepts are similar concepts, no articles directly comparing behavioural rebound and crowding-out mechanisms have been identified.

One reason for the inconsistent statements regarding the interconnectedness of behavioural rebound, negative spillover, and crowding-out mechanisms could be found in the observation that studies frequently build their conclusions upon traditional definitions. For instance, some studies build further upon the idea that compared to spillovers, rebound effects are based on a change in economic values rather than a change in preferences (e.g., Nilsson et al., 2017; Sorrell et al., 2020) and occur on a macro rather than a micro level (e.g., Nash et al., 2017; Truelove et al., 2014). Nevertheless, an increasing number of studies highlights the importance of the non-economic and micro-level factors involved in rebound mechanisms (section 1.1.1.). Similarly, an underlying argument has been that spillover mechanisms explain indirect effects, whereas rebound mechanisms explain both direct and indirect effects (e.g., Dütschke et al., 2018; Santarius and Soland, 2018). However, research has illustrated that spillover mechanisms can explain direct effects too, for instance, when a pro-environmental behaviour leads to less of the same pro-environmental behaviour on a later moment in time (i.e., temporary spillover) (Nilsson et al., 2017).

Furthermore, the conceptual differences between the mechanisms could be attributed to their origins in distinct disciplines. As previously noted, research on behavioural rebound effects has its roots in the energy and economics literature (e.g., Jevons, 2023), whereas spillover effects more recently emerged from psychological and marketing research (e.g., Bettels and Wiedmann, 2019; Nilsson et al., 2017) and crowding-out effects stem from a focus on policy interventions (e.g., Titmuss, 1971). While this disciplinary distinction helps explaining the variation in existing terminology, it should not justify continued conceptual ambiguity as an interdisciplinary perspective including streamlined terminology is essential for advancing rebound effect understanding (Andrew and Pigosso, 2024; Bracken and Oughton, 2006).

To conclude, the prevailing ambiguity and lack of conceptual precision in distinguishing behavioural rebound, negative spillover, and crowding-out mechanisms impedes the interpretation of research findings, as researchers frequently fail to specificy which mechanism is under investigation. The consequential challenges in interpretation and integration may slow down the progress of related research, which is alarming given the current need for a better understanding of the behavioural mechanisms undermining environmental sustainability efforts (Bracken and Oughton, 2006; Pischke et al., 2017; Van der Loo and Pigosso, 2024).

1.2. Study scope and outline

This study aims to untie the knot in the existing understanding of behavioural rebound, negative spillover, and crowding-out mechanisms, with a particular focus on (1) behavioural mechanisms (2) explaining an offsetting effect (3) in the context of environmental sustainability (Fig. 1).

Firstly, a behavioural mechanism is defined as a clear structure of causal processes where a stimulus (i.e., an intervention) interfaces with an actor (i.e., person or organisation) leading to a psychological (i.e., cognitive, can be irrational/unconscious) motivational change causing a

physical behavioural change (Fig. 2., adopted from Biely (2022) and Hedström and Ylikoski (2010)). Whereas spillover and crowding effects are regularly explained by behavioural mechanisms, this is relatively uncommon for rebound effects (Santarius and Soland, 2018; Van der Loo and Pigosso, 2024). To reflect the distinction, the term *behavioural rebound* is employed in this study to refer specifically to rebound effects caused by behavioural, rather than for instance economic (i.e., focus on utility maximisation) or social (i.e., focus on interacting actors), mechanisms.

Secondly, the offsetting scope criterion refers to a counteracting effect that decreases the potential positive impact. Reinforcing effects, referring to a strengthening effect that increases the potential positive impact, are therefore outside the scope of this study. Since the concept of spillover and crowding frequently encompass reinforcing effects, this study clarifies this distinction by referring to *negative-spillover* and *crowding-out*. Lastly, environmental sustainability in this study is expressed in terms of resource consumption. Fig. 2 displays how a behavioural mechanism can affect resource consumption, either directly (e.g., increased energy consumption) or indirectly (e.g., decreased lifetime of a car).

This paper comprehensively compares behavioural rebound, negative spillover, and crowding-out at a mechanism level. Building upon the findings from a systematic literature review (section 2.) the mechanisms are compared on a structural and contextual level (section 3.). Recommendations for interpreting and integrating insights from behavioural rebound, negative spillover, and crowding-out research are presented (sections 4 & 5), supporting future research

2. Methodology

The methodology of this study is divided into two stages: the systematic literature review (section 2.1.) and the content analysis (section 2.2.).

2.1. Literature review procedure

A systematic literature review, following the guidelines of Biolchini et al. (2005), is employed to obtain an overview of the behavioural rebound, negative spillover, and crowding-out mechanisms underlying rebound effects. The search string consists of three keywords and their synonyms: (1) 'rebound', 'spillover', or 'crowding' and (2) 'behaviour' in the title, and (3) 'sustainability' in the title, abstract, or keywords. The search string is displayed in Table 1.

Publications with keywords referring to topics including chemistry (e.g., nano, aerosol, antibody), medicine (e.g., cancer, pain, medic), geography (e.g., earthquake, geo, volcano), finance (e.g., stock market, liquidity, volatility), and humanities (e.g., anthropology, family, racism), are not included in the search due to their limited relevance to the study and low likelihood of examining behavioural mechanisms in an environmental sustainability context.

Scopus and Web of Science are used as scientific databases because of their extensive journal coverage and relevance to behavioural rebound, negative spillover, and crowding-out studies, encompassing a broad range of disciplines within social as well as interdisciplinary sciences (Ballew, 2009; Falagas et al., 2008). The search results in 1028 studies and was carried out in August 2025. The filtering process is presented in Fig. 3. Relevant studies are selected with the aid of three filters: (1) the article title, keywords and abstract; (2) introduction and conclusion; and finally (3) the full text. The inclusion criteria are in line with the scope of this paper (section 2.1.):

- Applying the concept of rebound, spillover, or crowding
- Elaborating on at least one behavioural mechanism
- Explaining an offset of potential positive impact
- Focusing on the environmental sustainability context

After the filtering process and exclusion of duplicates, non-English and not accessible articles, 48 studies are selected for the final analyses. Through backward snowballing, 4 additional studies that meet the inclusion criteria and provide a novel mechanism explanation are included in the review.

The total of 52 selected studies is further categorized into 15 rebound, 20 spillover and 17 crowding articles. Whenever studies mention multiple concepts, the concept used to define the relevant behavioural mechanism determines the categorisation of the study.

In total, 29 behavioural rebound, 30 negative spillover, and 20 crowding-out mechanisms are extracted from the articles and used for the analysis. The 51 mechanisms provide 17 distinct explanations for the observed offset of sustainability gains (section 3.1.).

Table 1
Overview of the search string.

	Search string
Keyword clusters	(TITLE ((“unintended consequence*" OR “backfire*" OR “crowd* out” OR “Jevons Paradox” OR “rebound*" OR “side effect*" OR “policy resistan*" OR “unexpected consequence*" OR “second-order effect*" OR “boomerang effect*" OR “spillover*" OR “efficiency paradox*" OR “perverse effect*" OR “indirect environment*" OR “ripple effect*" OR “nudg* out")) AND TITLE ((“behav*" OR “psycholog*" OR “social*" OR “cognitive*" OR “nudg*" OR “irrational*" OR “heuristic*" OR “bias” OR “moral*" OR “norm*" OR “prospect*" OR “dual processing” OR “neuro*" OR “individual*" OR “consumer*" OR “producer*")) AND TITLE-ABS-KEY ((“sustain*" OR “environment*" OR “circular*" OR “ecolog*" OR “climate*" OR “energ*" OR “life cycle*" OR “LCA” OR “green*" OR “nature” OR “dematerial*" OR “sharing econ*" OR “low-carbon” OR “sufficien*" OR “reuse” OR “efficien*" OR “effectiv*" OR “biodivers*")))
Excluded keywords	AND NOT TITLE-ABS-KEY (“antenna” OR “velocity” OR “racial” OR “workplace” OR “liquidity” OR “volatility” OR “geo” OR “anthrop” OR “gender” OR “crimin*" OR “stock market” OR “volcan*" OR “family” OR “race” OR “earthquake” OR “covid” OR “hydro*" OR “bact*" OR “cancer*" OR “pain*" OR “antibod*" OR “vegetat*" OR “nano*" OR “HIV” OR “vaccin*" OR “rock” OR “pharma*" OR “drug*" OR “steel*" OR “medic*" OR “racism*" OR “therm*" OR “photo*" OR “aerosol*")

2.2. Content analysis

The analysis of the identified mechanisms is divided into two steps. Firstly, behavioural rebound, negative spillover, and crowding-out mechanisms are systematically compared on a structural level to provide a comparison based on the causal elements included in the mechanisms (section 3.2.). Secondly, the mechanisms of each concept are systematically compared on a contextual level to provide a comparison in terms of analysed applications (section 3.3.). Both, the structural and

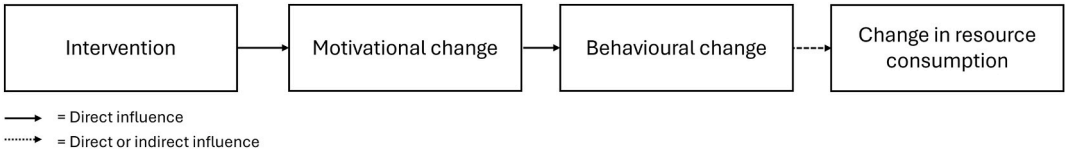


Fig. 2. Illustration of the causal processes within the behavioural mechanism.

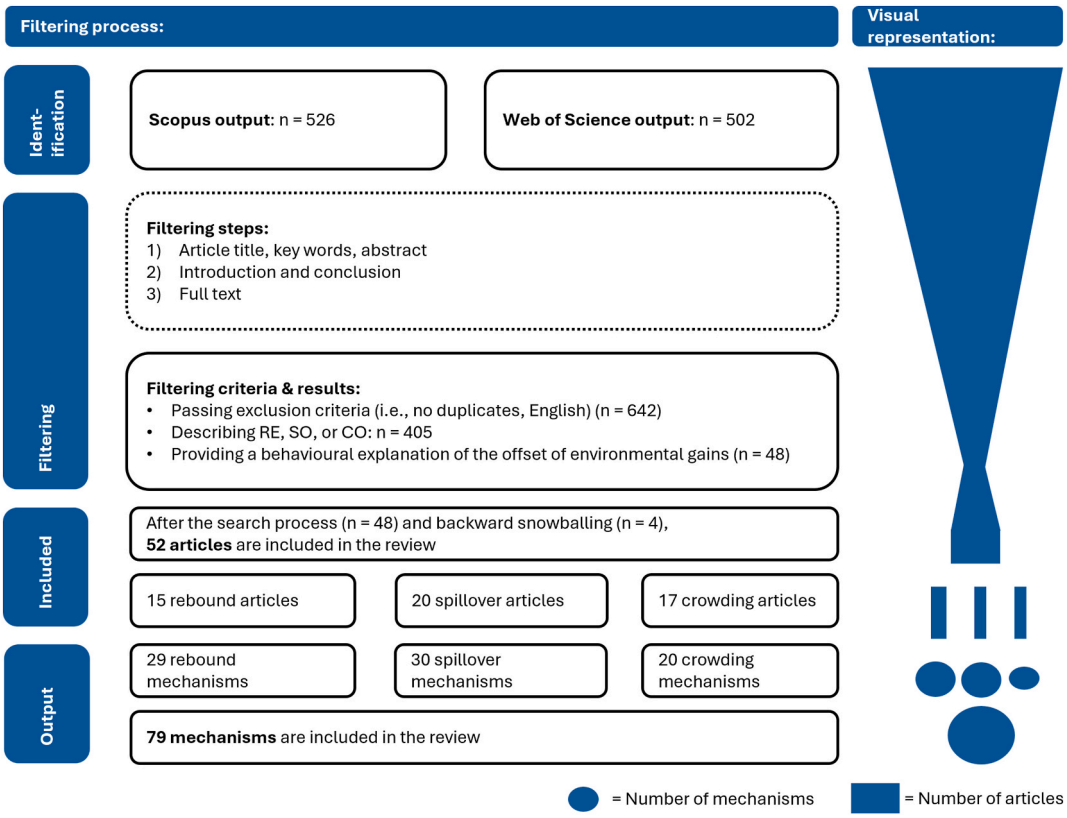


Fig. 3. A flow diagram of the systematic literature review, adapted from Biolchini et al. (2005)

Table 2
An overview of the units of analysis and related characteristics.

Unit of analysis	Description	Inspired by	Level of analysis	Mechanism part
Presence of external intervention	A mechanism can be caused by: (i) an external intervention (i.e., a purposeful action by a human agent to create change (Midgley, 2003) (e.g., a nudge, policy, or technology)); or (ii) a primary behaviour (Kumar et al., 2023) (e.g., eating vegetarian).	Austin et al., 2011; Kumar et al., 2023	Structure	Input
Intervention characteristic	External interventions can have different characteristics: (i) interventions targeting **autonomous** (i.e., intrinsic) motivation imply that individuals act voluntarily (e.g., an eco-label); (ii) interventions targeting **controlled** (i.e., extrinsic) motivation imply that individuals act due to external incentives, pressure or control (e.g., a ban or pricing mechanism) (Geiger et al., 2021; Ryan and Deci, 1985); and (iii) **technological** intervention are characterised by a change in a product or service without directly targeting at a motivational or behavioural change (e.g., energy efficient product design). If no external intervention is mentioned before the behaviour change, the intervention characteristic is documented as a primary behavioural change.	Geiger et al. (2021); Nilsson et al. (2017); Truelove et al., 2014	Context	Input
Number of behavioural changes	Minimum of physical behavioural changes (e.g., eating vegetarian, driving an electrical car) required for the mechanism to occur.	–	Structure	Overall mechanism
Explained effect type	The addressed effect type is classified as: (i) "direct" when the mechanism explains a change related to the targeted behaviour, product, product/service system, or socio-technical system; (ii) "indirect" when it considers a change related to a different behaviour, product, product/service system, or socio-technical system than targeted (Nilsson et al., 2017; Sorrell et al., 2020).	Dütschke et al. (2018); Nilsson et al. (2017); Santarius and Soland, 2018	Context	Overall mechanism
Identified end-point	The end-point of a mechanism is categorized as: (i) a change in motivation (e.g., support, norms, intrinsic motivation); (ii) a change in behaviour (e.g., recycling behaviour, turning down the heating); or (iii) a change in resource consumption (e.g., energy consumption, shortened lifetimes).	Dütschke et al. (2018); Nilsson et al. (2017), Truelove et al., 2014, Clot et al. (2022)	Structure	Output
Domain	The output or effect is analysed in different domains: (i) energy context (e.g. energy-efficient technology), (ii) nature conservation (e.g., forest management), or (iii) sustainability in general (e.g., changed pro-environmental attitudes). Whenever multiple domains are described (e.g., energy, material, and water-usage), the general domain is assigned.	Geiger et al. (2021); Nilsson et al. (2017)	Context	Output

contextual level analyses focus on similarities and differences in terms of the overall mechanism, input (i.e., starting point), and output (i.e., end point) characteristics.

Table 2 provides an overview of all units of analysis included in this study. The units of analysis have been selected based on cues from previous studies. The exception is the number of behavioural changes, which has been inductively added by the authors of this study. Data on the units of analysis is directly derived from the described mechanisms and complemented by general information coming from the selected papers.

Additionally, Fig. 4 displays schematically to which part of the mechanism each selected unit of analysis refers, and whether they concern a structural or contextual level.

The vertical dotted lines in Fig. 4 aim to illustrate that a characteristic is only categorized as an output level characteristic when it is the final end-point in the mechanism description. For example, when a motivational change is mentioned along with a subsequent behavioural change or change in resource consumption, the motivational change is not defined as the final output.

The overview of the different behavioural rebound, negative spillover, and crowding-out mechanisms and the structural and contextual comparison are used to derive recommendations for future studies, ultimately facilitating research on the often-neglected behavioural mechanisms underlying offsetting effects in an environmental sustainability context.

3. Results

The results of this study are presented in three subsections. First, the identified behavioural rebound, negative spillover, and crowding-out mechanisms are presented (section 3.1.), after which the comparison on a structural level is presented (section 3.2.), followed by the similarities and differences on a contextual level (section 3.3.).

3.1. Identified behavioural rebound, negative spillover, and crowding-out mechanisms

In total, 29 behavioural rebound, 30 negative spillover, and 20 crowding-out mechanisms have been extracted from the 52 reviewed articles. The identified mechanisms provide 17 distinct behavioural explanations for the occurrence of rebound effects, which are presented in Table 3. Whenever multiple articles describe the same mechanism, the given definitions are merged into one.

Some of the mechanisms have been identified by more than one term. Moral licensing (i.e., moral credentials), for example, has been explained as a behavioural rebound, negative spillover, and crowding-out mechanism. Nevertheless, each concept has identified mechanisms that have not been identified by any of the other concepts yet:

- Behavioural rebound is the only concept to have identified the temporal discounting; response efficacy; attitude; mental accounting; and personal/social norms mechanism.
- Negative spillover is the only concept to have identified the single action bias mechanism.
- Crowding-out is the only concept explaining control aversion and false hope

While the difference in the number of uniquely identified mechanisms might be the result of a difference in the number of studies conducted on each concept, it could also imply that the concept of behavioural rebound encompasses a larger scope of mechanisms compared to negative spillover and crowding-out mechanisms.

Furthermore, most behavioural rebound (41.4 %, n = 12) and negative spillover mechanisms (60.0 %, n = 18) describe moral licensing (i.e., moral credits), while crowding-out mechanisms predominantly refer to motivational crowding (35.0 %, n = 7). Although eleven, five, and nine alternative explanations have been identified for behavioural rebound, negative spillover, and crowding-out respectively, none have been examined in more than four publications. The relative popularity of moral licensing and motivational crowding may be attributed to three factors. Firstly, they may represent the most significant behavioural explanations for the offset of environmental benefits. However, the lack of an integrated empirical overview limits the ability to statistically compare effect sizes and test this assumption. Secondly, their prominence may stem from early identification in general psychology (e.g., Festinger, 1957) and health behaviour literature (e.g., Titmuss, 1971), which likely facilitated their adoption in rebound, spillover, and crowding research. Lastly, their frequent citation could be the result of a disproportionate research focus, driven by familiarity rather than empirical robustness or theoretical establishment.

By combining the identified mechanism elements for the three concepts, two pathways illustrating how environmental gains can be offset due to behavioural mechanisms are identified.

1. Pathway A involves an intervention, which leads to a motivational change and a corresponding behavioural change, eventually increasing resource consumption. An example of this pathway is the design of a fuel-efficient car (i.e., intervention), giving users of the car the perception that engineers are better able to solve environmental problems than themselves (i.e., diffusion of responsibility mechanism), decreasing user's motivation to drive sustainably (= motivation (I)), subsequently increasing the driven distances (= behaviour (I)), which ultimately increases the fuel consumption (i.e., resource consumption).
2. Pathway B involves an intervention that triggers an initial motivational and behavioural change, but only increases the resource consumption through a changed subsequent motivation and corresponding subsequent behaviour. An example of this pathway is the design of a fuel-efficient car (= intervention), which motivates a user

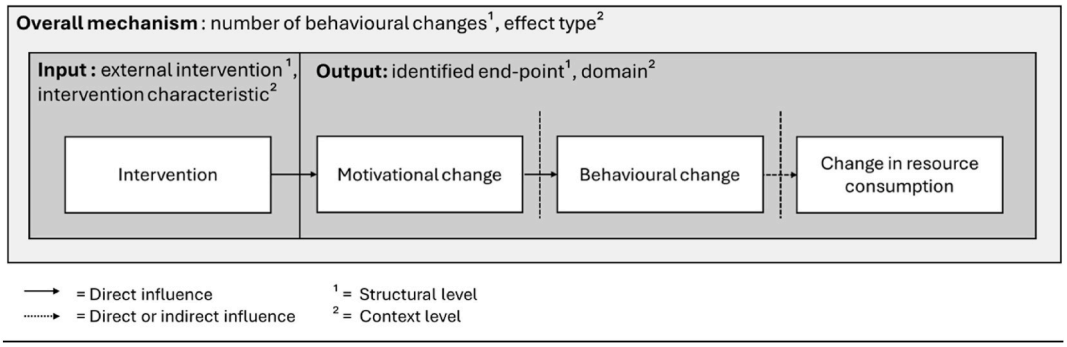


Fig. 4. A schematic representation of the selected units of analysis.

Table 3
An overview of the identified behavioural rebound (RE), negative spillover (SO), and crowding-out (CO) mechanisms.

Mechanism title	Definition	Used concept and (number of) related references
Moral licensing (i.e., moral credits)	Performing an initial morally virtuous behaviour (e.g., driving a more efficient car) can create the perception of moral entitlement to subsequent morally questionable behaviour (e.g., driving the car more).	RE (12): Barkemeyer et al. (2023); Dorner (2019); Dütschke et al. (2018); Dreijerink et al. (2023); Exadaktylos & van den Bergh (2021); Kerner and Brudermann (2021); Mathex et al. (2025); Rabaa et al. (2024); Reimers et al. (2021), 2022; Santarius and Soland (2018); Seger et al. (2025) SO (18): Carrico (2021); Castro et al. (2021); Clot et al. (2022); Dreijerink et al. (2021); Ek (2018); Haggar et al. (2023); Isbanner et al. (2021); Lacasse (2016); Lacroix et al. (2022); León et al. (2024); Maki et al. (2019), Nash et al. (2017); Nilsson et al. (2017); Sorrell et al. (2020); Spaccatini et al. (2023); Truelove et al. (2014); Verfuerth and Gregory-Smith (2018); Wang and Ma (2025)
Moral licensing (i.e., moral credentials)	Engaging in a behaviour that is perceived as morally good will increase positive emotions (e.g., less guilt). Subsequently, these positive emotions will inhibit the influence of discomfort when conducting a subsequent perceived immoral behaviour.	RE (1): Kerner and Brudermann (2021) SO (4): Lacasse (2016); Nash et al. (2017); Nilsson et al. (2017); Verfuerth and Gregory-Smith (2018) CO (3): Lemken et al. (2024); Rode et al. (2015); Knook et al. (2022)
Motivational crowding	Enticements, often increasing extrinsic motivation, can undermine moral norms, attitude, policy support, or other forms of intrinsic motivation for moral behaviour.	CO (7): Ferrara and Missios, 2024; Graafland and de Bakker (2021); Grepperud (2007); Kim et al. (2025); Ling et al. (2023); Vorlaufer et al. (2023); Wollbrant et al. (2022) SO (1): Maki et al. (2019)
Need satisfaction	An activity, product, or service which harms psychologic and/or basic needs can lead to a compensating increase in resource consumption.	RE (2): Bączyk et al. (2024); Hofstetter et al. (2006) CO (1): Ezzine-de-Blas et al., 2019
Value changes	External incentives can shift actors' attention towards a focus on economic reasoning. In the long term, the focus on economic reasoning can negatively influence the attitudes towards moral behaviour.	RE (1): Otto et al. (2014) SO (1): Isbanner et al. (2021) CO (2): Chervier et al. (2019); Rode et al. (2015)
Personal & social norms	After conducting a moral behaviour, actors might themselves, or have the perception that relevant others (e.g., peers and families), have fewer norm related expectations to conduct subsequent moral behaviours.	RE (3): Peters and Dütschke (2016) [a]; Seebauer (2018)
Contribution ethics	Having the belief of having appropriately contributed to a moral good results in limited subsequent moral behaviour.	SO (2): Nash et al. (2017); Nash and Whitmarsh (2023) CO (1): Werfel (2017)
Identity change	Incentives can lead to the perception that one undertakes a moral behaviour due to external rather than voluntary reasons, limiting the moral identity and subsequent behaviour.	CO (3): Chervier et al. (2019); Rode et al. (2015) [a] SO (2):Isbanner et al. (2021); Van Der Werff and Steg (2018) RE (1): Otto et al. (2014)
Single action bias	Interventions can trigger the belief that a single action can resolve a complex issue, leading to limited further actions.	SO (3): Nash et al. (2017), Nash and Whitmarsh (2023); Truelove et al. (2014)
Diffusion of responsibility	A sustainability-oriented intervention can shift the perceived responsibility for protecting the environment to other agents (e.g., engineers, policy makers), which leads to a decreased preference for frugal purchase/use of that technology, or of other technologies.	RE (2): Santarius and Soland (2018); Dreijerink et al. (2023) CO (1): Naatu et al. (2022)
Information overload	An intervention requiring high levels of attention or cognitive load can, due to limited cognitive capacity, lead to misconceptions and inattention related to (other) moral behaviours.	RE (1): Exadaktylos & van den Bergh (2021) CO (1): Lemken et al. (2024)
Mental accounting	Mental accounting explains the different values a person can place on the same amount of money. This can lead to irresponsive re-spending of saved resources due to improved technologies.	RE (1): Exadaktylos & van den Bergh (2021)
Attitude	An intervention can affect the perceived consequences of a product or service (e.g., the use is perceived as less costly or harmful to the environment), leading to increased use of the product or service.	RE (1): Peters and Dütschke (2016)
Response efficacy	An intervention can alter the awareness of a pro-environm ental behaviour as being an effective action to mitigate a perceived problem and consequently decrease engagement in this behaviour.	RE (4): Dorner (2019); Mathex et al. (2025); Peters and Dütschke (2016); Santarius and Soland (2018)
Temporal discounting	The tendency to overvalue immediate costs and benefits compared with future ones can lead to time inconsistent choices and irresponsive re-spending of saved resources due to improved technologies.	RE (1): Exadaktylos & van den Bergh (2021)
Control aversion	Actors can experience frustration, unfairness, or distrust when having the feeling of being controlled, leading to less of the targeted behaviour.	CO (1): Rode et al. (2015)
False hope	An intervention can decrease support for substantive policy by providing false hope that problems can be tackled without imposing considerable costs.	CO (1): Hagmann et al. (2019)

[a] This source has described the mechanism by explaining two sub-mechanisms.

to drive more fuel-efficient (= motivation (I) and behaviour (I)), giving the user the perception of having received 'moral credits' (i.e., moral licensing mechanism), which could function as a motivation to spend the 'moral credits' by taking a flight vacation (= motivation (II) and behaviour (II)), ultimately increasing the greenhouse gas emissions (= resource consumption).

The two main pathways (i.e., pathway A and B) are represented in Fig. 5, which consolidates all links and elements mentioned in the identified behavioural rebound, negative spillover, and crowding-out mechanisms.

Whereas behavioural rebound, negative spillover, and crowding-out mechanisms all describe how potential environmental sustainability gains can be offset due to behavioural mechanisms, a comparison on the structural reveals how the concepts differ in terms of causal elements.

3.2. Comparison on a structural level

Even though overlap exists, behavioural rebound, negative spillover, and crowding-out mechanisms address different sub-pathways (Fig. 6, ie., pathway 1–8) of the previously identified main pathways (Fig. 5, ie., pathway A and B):

- Behavioural rebound mechanisms commonly explain how an intervention can affect a behaviour (pathway 1) or multiple behaviours (pathway 3) leading to a direct or indirect change in resource

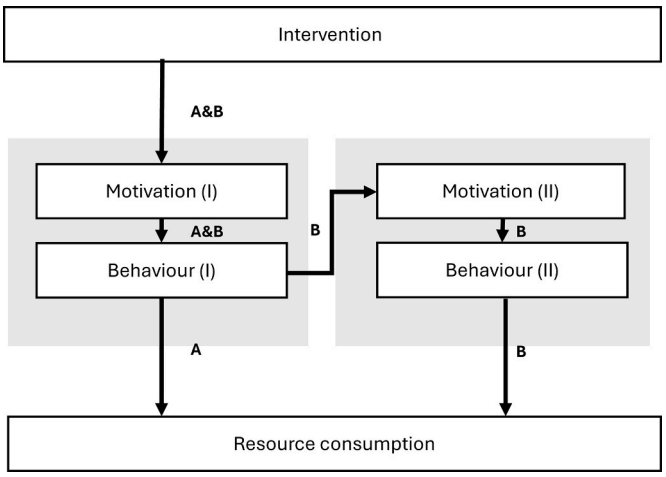


Fig. 5. An overview of all identified mechanism elements, links, and main pathway A & B.

consumption. Only three of the behavioural rebound mechanism address multiple motivational changes (pathway 2).

- Negative spillover mechanisms majorly explain how an intervention influences a behaviour which leads to a subsequent behavioural change (pathway 4). Three of the mechanisms identify a motivational instead of behavioural change as end-point (pathway 6), while merely two mechanisms highlight a change in resource consumption (pathway 3). Only two negative spillover mechanisms emphasize multiple motivational changes (pathway 5).
- Crowding-out mechanisms mostly explain how an intervention triggers a motivational change (pathway 7), and only rarely emphasize a behavioural change (pathway 8). None of the crowding-out mechanisms examines a change in resource consumption. Only one crowding-out mechanisms addresses how a behavioural change leads to a motivational change (pathway 6).

Furthermore, when comparing the different structures of behavioural rebound, negative spillover, and crowding-out mechanisms, it can be concluded that the three concepts differ in terms of input (e.g., presence of external intervention), output (e.g., identified end-point), and overall mechanism characteristics (e.g., number of behavioural changes). Table 4 displays three main results and implications.

First, compared to behavioural rebound and crowding-out mechanisms, negative spillover mechanisms most frequently do not involve an external intervention (46,7 %, n = 14). Nevertheless, some behavioural rebound and crowding-out mechanisms also lack an external intervention (20.7 %, n = 1 and 5.0 %, n = 1, *respectively*).

Second, the number of included behavioural changes differs across concepts. Behavioural rebound mechanisms involve one or two behavioural changes, negative spillover mechanisms majorly involve two behavioural changes, and crowding mechanisms involve none or one behavioural change. This result might explain behavioural rebound and negative spillover mechanisms frequently describe moral licensing, as this explanation requires two behavioural changes (e.g., collecting and re-spending of moral credits) to occur.

Lastly, a clear difference can be observed in terms of identified mechanism output, as behavioural rebound mechanisms refer to the output as a change in resource consumption (100 %, n = 29), negative spillover mechanisms majorly as a change in behaviour (93.3 %, n = 28), and crowding-out mechanisms as a change in motivation (95.0 %, n = 19). The focus of crowding-out mechanisms on motivational changes can explain why it often explains motivational crowding, which is based on changed intrinsic and extrinsic motivations.

In summary, regardless the structural differences, behavioural rebound, negative spillover, and crowding-out mechanisms show overlapping elements and links and could therefore be used to describe similar phenomena:

- Behavioural rebound and negative spillover mechanisms can describe the same phenomenon when it includes a behavioural change (I), leading to a subsequent behavioural change (II) which explains, either directly or indirectly, a change in resource consumption.
- Behavioural rebound and crowding-out mechanisms can describe the same phenomenon when it includes a triggered motivational change (I), leading to one or multiple subsequent behavioural changes (I or I&II), ultimately resulting in a change in resource consumption.
- Crowding-out and negative spillover mechanisms can describe the same phenomenon when a triggered motivational change (I) affects

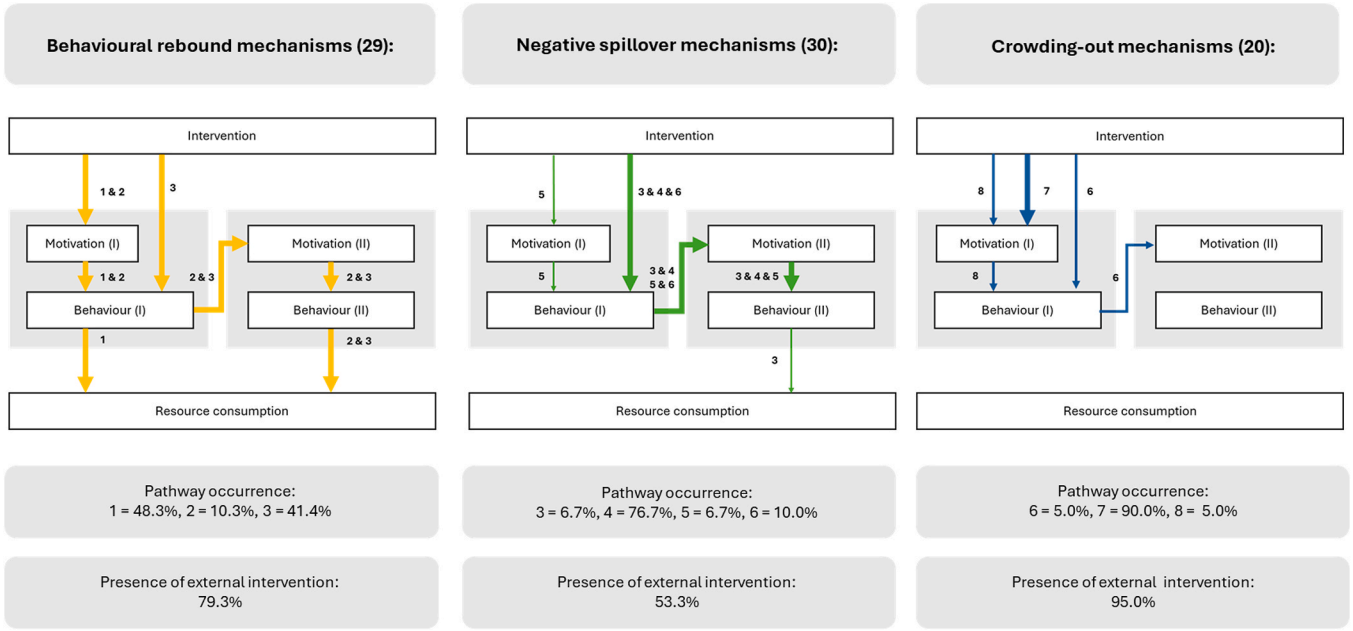


Fig. 6. Structural comparison of behavioural rebound, negative spillover, and crowding-out mechanisms.

Table 4
An overview of the structural differences between the different concepts.

Unit of analysis	Value	Behavioural rebound mechanism (n = 29)	Negative spillover mechanism (n = 30)	Crowding-out mechanism (n = 20)
Presence of external intervention	Present	79.3 %	53.3 %	95.0 %
	Not present	20.7 %	46.7 %	5.0 %
Number of behavioural changes	Zero	-	–	80.0 %
	One	44.8 %	3.3 %	20.0 %
	Two	55.2 %	96.7 %	-
Identified end-point	Motivation	-	-	95.0 %
	Behaviour	-	93.3 %	5.0 %
	Resource consumption	100%	6,7 %	-

the corresponding behaviour (I) as well as a subsequent behaviour (II).

An example of a phenomenon that can be described by a behavioural rebound, crowding-out, as well as negative spillover mechanism could involve a nudge which emphasizes the monetary benefits of eating vegetarian (=intervention), which increases the extrinsic motivation (= motivation (I)) to change an actor's diet from an animal-based into a vegetarian diet (= behaviour (I)). Simultaneously, this motivation and behaviour motivates the actor to behave cost-reducing in different situations (= motivation (II), leading to the consumption of less locally produced high-cost food (= behaviour (II), which consequently can increase the required energy for transporting food (= resource consumption).

Taken together, behavioural rebound, negative spillover, and crowding-out mechanisms have structural differences and should therefore not be used as synonyms. Nevertheless, when being aware of the differences and similarities, the three concepts could be integrated since overlapping phenomena are described.

3.3. Comparison on a contextual level

In addition to a comparison on a structural level, behavioural rebound, negative spillover, and crowding-out mechanisms are compared on a contextual level. First, the comparison based on the type of intervention over time is presented Fig. 7.

The results show that behavioural rebound mechanisms mostly examine technological interventions (65.5 %, n = 19), sometimes primary behaviours (20.7 %, n = 6), but rarely autonomous (6.9 %, n = 2) and controlling interventions (6.9 %, n = 2). The traditional focus on technological interventions is likely to be a result of the origins of rebound effects (e.g., Jevon's paradox), which describes the offsetting effect of (energy) efficient technologies.

Negative spillover mechanisms often examine primary behaviours (46.7 %, n = 14) and autonomous interventions (23.3 %, n = 7), but rarely controlling (13.3 %, n = 4) or technological interventions (16.7 %, n = 5). The focus on primary behaviours and autonomous interventions (e.g., nudge) directly targeting behaviours corresponds with the origin of negative spillover research in psychology and marketing research.

Lastly, crowding-out mechanisms mostly address controlling interventions (65.0 %, n = 13), sometimes autonomous interventions (30.0 % n = 6), but rarely a primary behaviour (5.0 % n = 1), and never a technological intervention. The focus on controlling interventions could reflect the existing perception that targeting a motivational change is more effective through a controlling than an autonomous intervention.

In short, behavioural rebound mechanisms mostly examine technological interventions whereas crowding-out mechanisms mostly examine controlling interventions. Among the three, negative spillover mechanisms encompass the widest range of intervention characteristics.

Additionally, the behavioural rebound, negative spillover, and crowding-out mechanisms are compared in terms of the explained effect type (Fig. 8).

Firstly, behavioural rebound mechanisms regularly explain direct effects (44.8 %, n = 13) and relatively often explain both, direct and indirect effects (31.0 %, n = 9). The lowest number of studies solely examining indirect effects (24.1 %, n = 7) could be explained by the fact that direct effects are often easier to measure than different effect types.

Negative spillover mechanisms often describe indirect effects (53.3 %, n = 16) or direct and indirect effects (46.6 %, n = 14) but are never used to explain direct effects alone. This result could reflect the traditional perception that a spillover effect concerns a change of one behaviour to another behaviour (i.e., behavioural spillover) rather than a spillover to the same behaviour in different contexts (i.e., contextual spillover) or moments in time (i.e., temporal spillover).

Lastly, crowding-out mechanisms often explain direct effects (70.0 %, n = 14), sometimes indirect effects (30.0 %, n = 6), but never the combination of direct and indirect effects. An explanation for the focus on direct effects could come from the observation that crowding-out mechanisms are often explained with the aid of motivational crowding, which creates a higher level of conflict when a controlling intervention decreases the intrinsic motivation for this behaviour (i.e., direct effect type) compared to another behaviour (i.e., indirect effect).

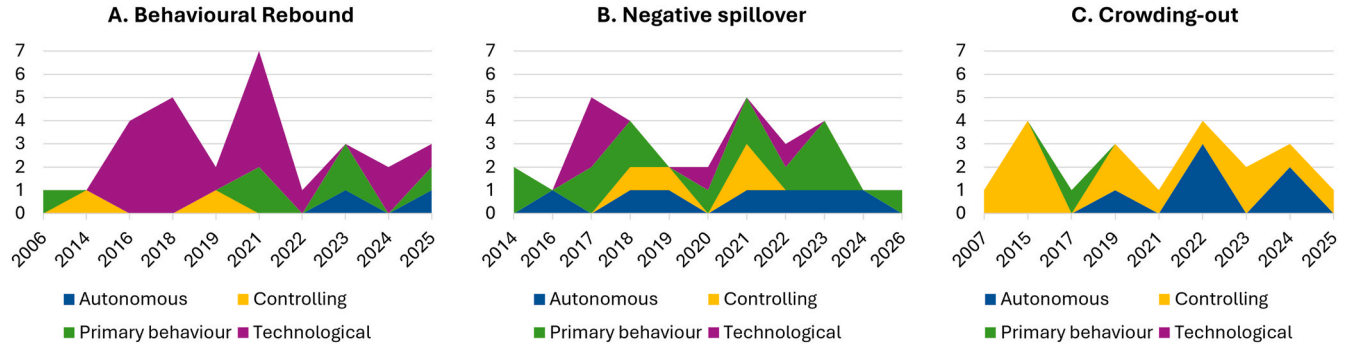


Fig. 7. A comparison of examined intervention characteristics per concept (i.e., behavioural rebound (A), negative spillover (B) and crowding-out (C) mechanisms).

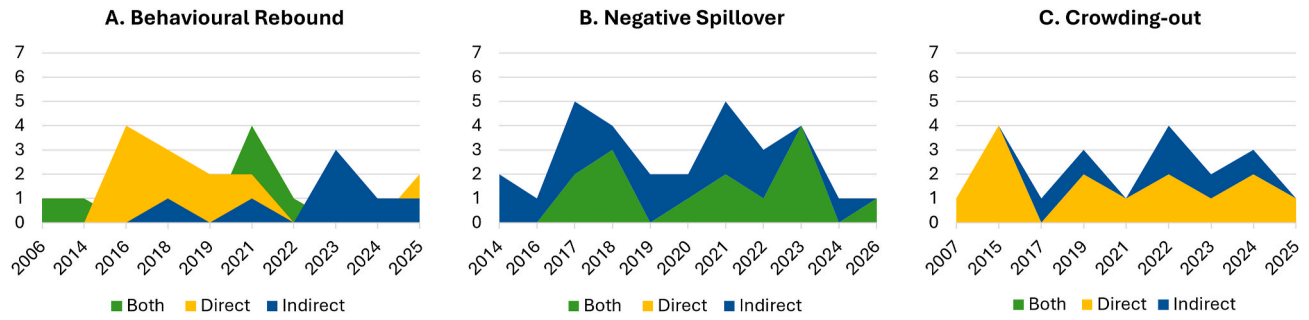


Fig. 8. A comparison of the explained effect types per concept (i.e., behavioural rebound (A), negative spillover (B) and crowding-out (C) mechanisms).

In sum, whereas negative spillover mechanisms mostly explain indirect effects, crowding-out mechanisms majorly explain direct effects. Behavioural rebound mechanisms explain direct and indirect effects in a more balanced manner.

Finally, behavioural rebound, negative spillover, and crowding-out mechanisms are compared in terms of examined domain (Fig. 9).

Firstly, it can be observed that behavioural rebound mechanisms mostly address the energy context (51.7 %, n = 15), sometimes the general sustainability context (37.9 %, n = 11), and rarely another specific sustainability context (10.3 %, n = 3). Even though slightly more variety in terms of examined context can be observed recently, the focus on the energy context in behavioural rebound research is aligns with the overrepresentation of the energy context in rebound research in general.

Additionally, most of the negative spillover mechanisms explain a change in a general sustainability context (90.0 %, n = 27) and sometimes the energy (3.33 %, n = 1) or other specific sustainability contexts (e.g., recycling) (6.7 %, n = 2). This result can potentially be explained by the observation that negative spillover mechanisms are often defined in terms of pro-environmental behaviour, encompassing multiple behaviours related to environmental sustainability rather than solely a specific context.

The output of crowding-out mechanisms is predominantly examined in a general sustainability context (50.0 %, n = 10) or a conservation context (30.0 %, n = 6), and less frequently in a specific other context (20.0 %, n = 4). One reason for the relative frequent focus on conservation may be that crowding-out mechanisms often study the effects of controlling policies, which tend to target specific areas like nature conservation rather than broader sustainability issues.

In short, behavioural rebound mechanisms are the only mechanisms showing a bias towards the energy context, whereas negative spillover mechanisms often examine the sustainability context in general, and crowding-out mechanisms additionally consider the slightly more specific conservation context.

The application level comparison illustrates that behavioural rebound, negative spillover, and crowding-out mechanisms do not only differ from a structural perspective but additionally display different trends in terms of analysed contexts.

4. Discussion

Existing research inconsistently refers to behavioural rebound, negative spillover, and crowding-out mechanisms, constraining the interpretation and integration of research findings and limiting the understanding of the often-neglected behavioural mechanisms underlying offsetting effects in the field of environmental sustainability. In this section, a discussion is provided in relation to the interpretation of findings in comparison with existing research (section 4.1.) and recommendations for future studies (section 4.2.).

4.1. Interpretation of findings and comparison existing research

Existing research has briefly examined the interconnectedness of solely behavioural rebound and negative spillover mechanisms (e.g., Exadaktylos & van den Bergh, 2021; Reimers et al., 2021, 2022), or negative spillover and crowding-out mechanisms (e.g., Clot et al., 2022; Truelove et al., 2014). The results of the output analysis on a structural level imply that:

(1) behavioural rebound mechanisms explain a change in resource consumption;
(2) negative spillover mechanisms explain a change in behaviour; and
(3) crowding-out mechanisms explain a change in motivation.

In line with the structure of a behavioural mechanism (Fig. 2.), this suggests that crowding-out mechanisms can lead to negative spillover mechanisms, which, in turn, can lead to behavioural rebound mechanisms. These findings add to the state-of-the art by pointing out the sequential connection of behavioural rebound, negative spillover, and crowding-out mechanisms, implying that the three concepts cannot be interchangeably used as synonyms.

Moreover, the results of the analysis at the structural level show that the number of included behavioural changes differs per mechanism (i.e., zero or one for crowding-out, one or two for behavioural rebound, and two for negative spillover). This indicates that negative spillover mechanisms could be seen as a type of behavioural rebound mechanism.

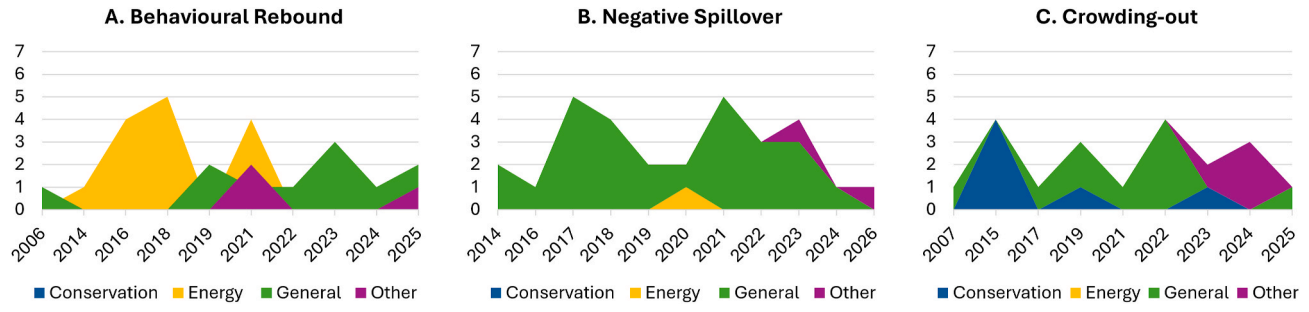


Fig. 9. Comparison of the examined domains per concept (i.e., behavioural rebound (A), negative spillover (B) and crowding-out (C) mechanisms).

Based on this result, all negative spillover mechanisms can be referred to as behavioural rebound mechanisms, whereas not all behavioural rebound mechanisms can be identified as negative spillover mechanisms. This finding contradicts previous studies arguing that rebounds are a type of spillover mechanisms (e.g., Clot et al., 2022; Nilsson et al., 2017). The contradiction could be explained by a difference in underlying arguments (e.g., structural rather than purely contextual) or by used mechanism definitions (e.g., modern rather than traditional).

The structural level results show that the presence of an external intervention within the mechanism is more common for behavioural rebound (79.3 %) and crowding-out (95.0 %) mechanisms than for negative spillover mechanisms (53.3 %), suggesting that a mechanism can be caused by a primary behavioural change. While the involvement of interventions is an ongoing discussion within negative spillover research (Kumar et al., 2023), this topic is yet to be addressed in behavioural rebound and crowding-out research.

Finally, the found contextual differences between behavioural rebound, negative spillover, and crowding-out mechanisms are majorly in line with previous studies. For instance, behavioural rebound mechanisms often address technological interventions in an energy context, negative spillover mechanisms often examine indirect effects in a general sustainability context, and crowding-out mechanisms majorly involve controlling interventions explaining a direct effect. This indicates that behavioural rebound, negative spillover, and crowding-out mechanisms have different strengths and weaknesses, which can be taken into account when studying offsetting behavioural mechanisms in a specific context.

4.2. Recommendations for future studies

This study compares 79 mechanisms identified across 52 articles, encompassing behavioural rebound, negative spillover, and crowding-out mechanisms. The relatively limited number of relevant publications may be attributed to the novelty of this research area (Santarius and Soland, 2018; Van der Loo and Pigosso, 2024). Subsequent studies comparing behavioural rebound, negative spillover, and crowding-out mechanisms could build further upon this by including upcoming studies and expanding the focus to, for instance, economic mechanisms, reinforcing effects, or other fields such as socio-economic sustainability.

Generally, the results of this study can be translated into two main recommendations for behavioural rebound, negative spillover, or crowding-out researchers and practitioners to facilitate further understanding of the behavioural mechanisms underlying offsetting effects in the environmental sustainability context:

(1) **Enhance precision in the use of behavioural rebound, negative spillover, and crowding-out,** building upon the structural comparison of the three types of mechanisms (Fig. 6). The use of precise and consistent terminology will strengthen the understanding of the behavioural mechanisms underlying offsetting effects, clarifying what is being referred to when the concepts are mentioned, either in isolation or in combination.

(2) **Integrate findings from behavioural rebound, negative spillover, and crowding-out mechanisms.** Due to the differences in structural and contextual characteristics, the three types of mechanisms can be of added value to one another.
   a. Firstly, combining the structural insights of behavioural rebound, negative spillover, and crowding-out mechanisms provides a more comprehensive understanding of the causal chain of output reactions, and aid in addressing the intention-behaviour gap (Geiger et al., 2021).
   b. Secondly, integrating the three concepts from a contextual perspective can contribute to tackling specific concept-related research gaps:
      i. Negative spillover and crowding-out mechanisms add value to behavioural rebound mechanisms by providing insights that go beyond the energy context and technological interventions;
      ii. Negative spillover mechanisms can adopt insights from behavioural rebound and crowding-out mechanisms in terms of explaining direct effect types (i.e., an effect in the same behaviour, product, product/service system, or socio-technical system as targeted);
      iii. Crowding-out mechanisms can be inspired by the non-controlling interventions (i.e., individuals act voluntarily) of negative spillover and behavioural rebound mechanisms.
   c. Finally, precision should also be applied during the process of integrating findings by considering the identified and possible additional structural and contextual differences, ensuring an effective flow of information from one area to another.

In addition to enhancing the interpretation and integration of insights related to behavioural rebound, negative spillover, and crowding-out mechanisms, these two recommendations have the potential to enhance the recognition of the mechanisms in practice. Improved recognition and understanding of behavioural rebound, negative spillover, and crowding-out mechanisms by researchers, designers, and policy-makers aids the development of tailored prevention strategies which ultimately could reduce the offsetting effects and consequent negative environmental impacts.

5. Conclusions

Behavioural rebound, negative spillover, and crowding-out are often used intertwiningly, without agreement on their key similarities and differences. Through a systematic literature review, 79 relevant mechanisms have been identified, consisting of 29 behavioural rebound, 30 negative spillover, and 20 crowding-out mechanisms. All mechanisms have been compared on a structural and contextual level, while analysing input, output, and overall mechanism characteristics.

While behavioural rebound mechanisms regularly explain how a (technological) intervention leads to one or more behavioural changes (resulting in a direct or indirect change in resource (energy) consumption), negative spillover mechanisms describe how one behaviour leads to a (different) offsetting behaviour (in the general sustainability context), sometimes without involving an intervention. Finally, crowding-out mechanisms majorly describe how a (controlling) intervention leads to a change in motivation. This indicates that negative spillover mechanisms can be seen as a type of behavioural rebound mechanisms, and that crowding-out mechanisms could lead to negative spillover mechanisms, ultimately resulting in behavioural rebound mechanisms.

To further advance understanding of behavioural rebound, negative spillover, and crowding-out mechanisms, two main recommendations have been identified: (1) enhance precision in the use of behavioural rebound, negative spillover, and crowding-out mechanisms; and (2) integrate the insights coming from the different mechanisms to build further upon the strengths and weaknesses of each concept.

By untying the knot of behavioural rebound, negative spillover, and crowding-out mechanisms, this research provides a shared vocabulary to facilitate both the quality and quantity of studies exploring behavioural mechanisms underlying offsetting effects in the environmental sustainability context.

CRediT authorship contribution statement

Imke G.H. van der Loo: Writing – review & editing, Writing – original draft, Visualization, Validation, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Daniela C.A. Pigosso:** Writing – review & editing, Validation, Supervision, Project administration, Methodology, Investigation, Funding acquisition,

Formal analysis.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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